

Anshika Shukla

Edge AI · Computer Vision · Embedded Systems · AI/ML Engineering
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PROFESSIONAL SUMMARY

ECE undergraduate specializing in **Edge AI, computer vision, and embedded ML** — with a proven record of shipping 6+ production-grade AI systems: CNNs trained on 80K+ datasets, TensorFlow Lite inference pipelines achieving 95% accuracy on Raspberry Pi, and satellite-assisted emergency response models trained on 20K+ samples at 87%+ classification accuracy.

Proficient in end-to-end model deployment, MLOps, computer vision pipeline development, Flask/FastAPI backend architecture, and IoT system design. Certified by NASA ARSET, Google Cloud, Microsoft, and MathWorks.

2nd Runner-Up HACKVERSE (National, 30-hr) · Top 10 HACKCELESTIA (National) · Selected — Sanchar Saathi Initiative, Govt. of India.

KEY ACHIEVEMENTS

- **2nd Runner-Up — HACKVERSE (National, 30-hr):** Engineered SafeGaze with 95% detection accuracy across 100+ competing teams using deep learning-based content moderation.
- **Top 10 — HACKCELESTIA (National):** Architected SatLink, an edge-AI emergency response system integrating satellite communication with real-time hazard detection.
- **Selected — Sanchar Saathi Initiative (Govt. of India, DoT):** Chosen for national initiative on digital safety and telecommunications innovation.
- **Technical Workshop Lead:** Delivered 5+ embedded systems workshops to 200+ students; mentored 20+ juniors; improved lab participation by 35%.

TECHNICAL SKILLS

Core Expertise	Edge AI · Computer Vision · Embedded ML · Deep Learning · CNNs · Model Deployment · Model Quantization · Inference Optimization · MLOps · IoT Systems
Languages	Python · C/C++ · JavaScript · MATLAB · SQL · HTML5/CSS3
AI/ML Frameworks	TensorFlow · TensorFlow Lite · Keras · Scikit-learn · OpenCV · NumPy · Matplotlib · Hugging Face Transformers
Hardware & Systems	Raspberry Pi · Arduino Uno · Embedded Sensors and Modules · MQTT · WebSockets · YOLO Detection
APIs & Integration	Flask API · FastAPI · REST APIs · Twilio API · Hugging Face API · Google Cloud APIs
AI Productivity	Prompt Engineering · Vibe Coding · LLM-Assisted Development
Tools	Git · GitHub · VS Code · Jupyter Notebook · Streamlit · LaTeX · MS Office · Canva
Soft Skills	Leadership · Mentoring · Communication · Problem-Solving · Adaptability · Teamwork · Cross-Cultural Coordination

PROJECTS

SMART-GUARD — IoT-Enabled Smart Helmet Safety System	April 2026
Arduino IDE · C++ · IoT · MPU6050 · GSM Module · Twilio API	
• Architected IoT-enabled smart helmet integrating Arduino Uno microcontroller with multi-sensor fusion (MPU6050, IR, MQ3, UV) for real-time accident detection and rider safety monitoring across 4 safety parameters. • Developed GSM/Twilio API integration achieving sub-2-second emergency alert delivery to predefined contacts with GPS-based location tracking; real-time sensor data streaming at 10Hz.	
AI Customer Support Agent — Intelligent Support Automation System	April 2026
Python · FastAPI · Hugging Face Transformers · NLP · Streamlit	
• Engineered real-time AI support agent with multi-task learning (sentiment analysis, intent detection, churn-risk scoring) across 5 emotion categories, automating escalation decisions across 3+ channels. • Achieved 85%+ intent detection accuracy; reduced manual escalation overhead by 30% and improved first-contact resolution by 25% through intelligent routing and personalized responses. • Architected scalable FastAPI backend with Streamlit dashboard; achieved sub-500ms end-to-end latency supporting concurrent request handling at 100+ req/sec.	
SatLink — Edge-AI Emergency Response System	Jan 2026
Python · FastAPI · TensorFlow · Scikit-learn · Edge AI	
• Architected AI-powered emergency response platform for low-connectivity disaster zones; trained ensemble hazard prediction models on 20K+ samples achieving 87%+ classification accuracy across 4 hazard categories. • Integrated edge-based computer vision pipeline with thermal sensing; real-time inference at 12 FPS on Raspberry Pi with 150ms latency; automatic alert generation with satellite/UAV data relay. • Deployed lightweight model achieving 95% accuracy retention post-quantization; designed architecture for sub-10-second data relay in zero-connectivity environments using cached inference.	

Real-Time Satellite Safety and Astronaut Health Dashboard

Oct 2025

YOLO-Tiny · MQTT · WebSockets · Edge AI · Computer Vision

- Deployed YOLO-Tiny real-time object detection pipeline for space debris and anomaly detection; achieved 15+ FPS inference throughput on Jetson Nano with 95%+ accuracy retention post-quantization.
- Engineered MQTT/WebSockets protocol stack with 180ms end-to-end latency; processed 1000+ telemetry updates/sec with automatic anomaly scoring and threshold-based alert triggering.
- Built live dashboard processing real-time sensor feeds; reduced manual monitoring workload by 60% through automated anomaly detection and escalation for mission-critical events.

SignEase — Real-Time Sign Language Recognition

Sept 2025

Python · TensorFlow Lite · OpenCV · Raspberry Pi · gTTS

- Trained CNN on 80K+ ASL images; achieved 95% accuracy; post-quantization model size reduced from 45MB to 8.5MB (81% compression) deployed on Raspberry Pi with 94% accuracy retention.
- Engineered model quantization and inference optimization achieving <100ms latency per gesture; applied data augmentation (rotation, flipping, brightness) improving robustness by 18% on held-out test set.
- Integrated gTTS speech synthesis enabling accessible real-time ASL-to-speech communication; deployed on resource-constrained edge hardware with <2GB RAM footprint.

Satellite Orbit Calculator

Aug 2025

C++ · Mathematical Modeling · MATLAB

- Implemented Kepler's Third Law and full orbital mechanics equations in C++ for accurate orbital period computation across 5+ satellite orbit types.
- Optimized computational efficiency achieving 3x faster parameter calculation vs. baseline MATLAB prototype for real-time orbital analysis.

SafeGaze — AI Content Moderation System

March 2025

Python · Flask · DeepFace · Custom NSFW Models · Deep Learning

- Engineered custom deep learning model achieving 95% detection accuracy; integrated DeepFace for facial analysis reducing false positives by 40% vs. baseline classifier across 10K+ test images.
- Built Flask API dashboard for real-time content moderation; processed 500+ media items/minute with sub-300ms inference latency, automated flagging, and confidence scoring for review triage.

EXPERIENCE

Technical Workshop Lead / Core Member — SAMARTHYA Electronics Club

Sept 2024 – Present

Banasthali Vidyapith, Jaipur

- Designed and delivered 5+ technical workshops on embedded systems, microcontroller programming, and IoT to 200+ students; improved lab participation by 35% and hands-on project completion by 40%.
- Coordinated 15+ core team members on technical events, national hackathons, and inter-college competitions; mentored 20+ junior students in circuit design and embedded project development.

Technical Head — Kshitij, French Cultural Club

Sept 2025 – Present

Banasthali Vidyapith, Jaipur

- Architected end-to-end technical workflow for bilingual club workshops and cultural programs — AV configuration, venue setup, digital registration systems, and run-of-show coordination for 50+ attendees.
- Led on-site technical execution of cross-cultural workshop with foreign exchange students; managed AV setup, bilingual content delivery, and real-time logistics across 3 concurrent activity tracks.

EDUCATION

B.Tech — Electronics and Communication Engineering

Banasthali Vidyapith, Jaipur | July 2023 – May 2027 | CGPA: 7.25 / 10

Research and Academic Contributions

- Research Paper: Advancements in Brain-Computer Interfaces for Communication Disorders
- Seminar: Satellite-based Narrowband IoT for Remote Connectivity
- Poster Presentation — WACIIS 2025 International Conference

CERTIFICATIONS

- Drone Certification — Department of AI, Banasthali Vidyapith
- Career Essentials in Generative AI — Microsoft and LinkedIn
- NASA ARSET — Applied Remote Sensing Training Program
- Building Data and AI Solutions on Google Cloud — 365 Data Science (Score: 100%, March 2026)
- MATLAB Onramp — MathWorks
- Diploma in French Language — Banasthali Vidyapith

Languages: English (Proficient) · Hindi (Native) · French (Diploma — Banasthali Vidyapith)